

Memory Circuits & Systems

Exercise 2

6-to-64 Row Decoder Design

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6-to-64 row decoder

- Design a **6-to-64 row decoder** for a SRAM array no matter which logic style. The I/O descriptions of the row decoder design are listed in Table 1. All **the input data must be connected to a buffer** so that the inputs won't have infinite driving ability.
 - Use a **high-level program**, such as C/C++, Python, Verilog, or generative AI models, to **generate the netlist file for HSPICE simulation**. Please provide the generation log if AI models are used.

Table 1. I/O description for register file

Pin Name	I/O	Description
Clock	Input	Clock input
Address[5:0]	Input	Read/Write Address
Wordline[63:0]	Output	64 Word-lines

Specifications of row decoder

- The frequency (data sampling rate) should reach to 1GHz (50% duty cycle).
- Address signals are changed at the falling edge of clock.
- The wordline would be asserted when clock is 1.
- The wordline capacitances depend on gate capacitances of the pass transistors in memory cells and wire loading.
 - ◆ The loading of each wordline is based on 1:2:2 SRAM and 7nm wire model. (64 SRAM cells on a row)
 - ◆ Please write down the calculation process in the report, not just the final number of the wire loading.
- The rise time and fall time of inputs are 0.1ns respectively.

Report requirements

■ Waveforms:

- ◆ Input signals for decoder
- ◆ Correct outputs of 6-to-64 Row Decoder

■ Architecture of 6-to-64 decoder:

- ◆ Provide the block diagram of your design
- ◆ Decoder design description

■ Estimation & Analysis:

- ◆ Wire Loading estimation process
- ◆ Delay and Power measurement and analysis of the decoder

■ Grading Policy:

- ◆ Functionality - 30 points
- ◆ Architecture – 10 points
- ◆ Wire Loading Estimation - 20 points
- ◆ Performance & Power Analysis - 30 points
- ◆ High-level Program - 10 points

Submission on E3 platform

- Please compress your **report & source codes** in a single **compressed file (.zip)** and upload this single file on E3 platform.
 - ◆ Report: **Report your analysis and results, including the architecture, performance and power consumption.**
- Naming rules of the compressed file
 - ◆ Upload file: **Ex2_#ID.zip** (including report & source code)
 - ◆ E.g. Ex2_409510021.zip
- Due date: 4/24 **23:55**